

QBUS3820 26s1 Final Practice

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W1 Introduction to Machine Learning

Q1. Short Answer Semester 1 2022 Final (modified)

A platform wants to automate high-value purchasing decisions from predicted asset prices. State two safeguards that should be built around the model before large-scale deployment, and explain how each reduces business risk.

Q2. MCQ Key: labelled data

A retailer wants to forecast demand for new stores using past store attributes and sales. Which description best matches supervised learning?

- (A) Grouping stores without an outcome variable.
- (B) Choosing an action only after receiving rewards.
- (C) Manually writing all decision rules.
- (D) Learning from input-output examples to predict future sales.

Q3. True/False Pitfall: deployment gap

A high validation score always proves that a model will perform well after deployment. Explain briefly.

Q4. Short Answer

Give one example each of structured and unstructured data in a business machine-learning project, and name a suitable model family for each.

Q5. Calculation Warning: cost asymmetry

A false positive costs 20 and a false negative costs 200. If a threshold change creates 30 extra false positives but prevents 5 false negatives, what is the net cost change?

Q6. Short Answer

Why is a simple baseline model useful even when the final model will probably be a neural network?

Q7. MCQ

Which choice is the clearest business target?

- (A) Improve analytics.
- (B) Predict whether a subscriber churns within 30 days.
- (C) Use AI.
- (D) Find insights.

Q8. Table/Diagram

Create a two-column table matching task type to output: regression, binary classification, multi-class classification.

Q9. Long Question

A bank wants to automate loan approvals using ML. List three non-model risks that must be controlled before deployment.

Q10. Long Question Pitfall: tail risk

A home-insurance firm wants an ML system to price risk automatically. Design a short deployment plan that includes the prediction target, validation evidence, uncertainty handling, and one monitoring trigger.



W2 Machine Learning Fundamentals

Q1. MCQ Semester 1 2023 Final (modified)

A classifier has very low training error but much worse validation error. Which diagnosis is most plausible?

- (A) The model is overfitting the training data.
- (B) The learning rate is necessarily zero.
- (C) The target variable is continuous.
- (D) The validation set has been used as training labels.

Q2. Calculation

A model has train RMSE 2.1 and validation RMSE 9.8. Another has train RMSE 5.4 and validation RMSE 6.0. Which is likely preferable and why?

Q3. MCQ Pitfall: accuracy trap

For a rare fraud class, which metric is least sufficient by itself?

- (A) Accuracy.
- (B) Recall.
- (C) Precision.
- (D) F1 score.

Q4. True/False

Ridge regularisation usually increases model flexibility because it allows larger coefficients.

Q5. Short Answer Key: zero recall

Explain the difference between model parameters and hyperparameters, giving one example of each.

Q6. Calculation

In a validation set of 1000 cases, 40 are fraud. A classifier predicts no fraud. Compute accuracy and explain the problem.

Q7. Short Answer Formula: log loss

When would log loss be more informative than accuracy?

Q8. MCQ

Which change most directly reduces variance?

- (A) Increase tree depth.
- (B) Tune on the test set.
- (C) Add regularisation.
- (D) Remove the validation set.

Q9. Long Question

A marketing model predicts campaign response probabilities. Describe how you would choose a decision threshold when contacting customers has a cost.

Q10. Long Question Strategy: cost threshold

A fraud model has 99 percent accuracy but misses most fraud cases. Explain why this is possible and propose a metric-threshold strategy tied to business cost.



W3 Machine Learning in Practice

Q1. Short Answer Semester 1 2022 Final (modified)

A team reports only accuracy for an imbalanced binary classification model. Explain why this may be misleading metrics or validation refinements that would give a more reliable model comparison.

Q2. MCQ Key: train-only fit

Which operation must be fitted on the training data only?

- (A) Counting rows in the whole dataset.
- (B) Stating the model objective.
- (C) Reading the business brief.
- (D) Standardisation means and variances.

Q3. True/False

Random search can be more efficient than grid search when only a few hyperparameters strongly affect performance.

Q4. Short Answer Pitfall: leakage

Name two signs that a feature may be leakage.

Q5. Calculation

A 5-fold CV reports AUC scores 0.76, 0.74, 0.79, 0.75, 0.76. Give the mean AUC.

Q6. Short Answer Strategy: pipeline CV

Why should preprocessing steps be inside a cross-validation pipeline?

Q7. MCQ

A model is selected after repeatedly checking the test set. Main risk?

- (A) Reduced training runtime.
- (B) Test overfitting.
- (C) More independent evidence.
- (D) Guaranteed deployment validity.

Q8. Table/Diagram

Make a compact table with three rows: high bias symptom, high variance symptom, distribution shift symptom.

Q9. Long Question

For a customer churn project, propose a validation split and explain how it avoids a misleading performance estimate.

Q10. Long Question Warning: test overfit

Design a leakage-safe model-selection protocol for churn prediction when customers have monthly records over several years.



W4 Training Machine Learning Models Part 1

Q1. Short Answer Semester 1 2023 Final (modified)

For regularised logistic regression, explain what the regularisation term is trying to achieve and how increasing the strength usually affects bias and variance.

Q2. MCQ Key: output-loss match

Which output/loss pair is natural for binary classification?

- (A) Softmax with MSE only.
- (B) Linear output with no loss.
- (C) Max pooling with hinge loss.
- (D) Sigmoid with binary cross-entropy.

Q3. True/False

Maximum likelihood estimation can often be written as minimising a negative log-likelihood.

Q4. Calculation Formula: sigma(0)

For logistic regression, if $x^T \beta = 0$, what is the predicted probability for class 1?

Q5. Short Answer

Why does standardising features help gradient-based training of regularised linear models?

Q6. Short Answer Pitfall: L2 not sparse

Compare L1 and L2 regularisation in terms of coefficient sparsity.

Q7. MCQ

In regularised ERM, what does a larger λ usually do?

- (A) Remove the loss term.
- (B) Strengthen the penalty.
- (C) Add more training data.
- (D) Force test leakage.

Q8. Calculation

Compute binary cross-entropy for $y = 1$, $\hat{p} = 0.8$ as an expression.

Q9. Long Question

A credit model must be interpretable and stable. Justify a regularised logistic regression baseline and state one limitation.

Q10. Long Question Strategy: log-loss choice

A regularised logistic regression and an unregularised one have similar training accuracy but different validation log loss. Explain how you would choose between them.



W5 Training Machine Learning Models Part 2

Q1. Short Answer Semester 1 2023 Final (modified)

Gradient descent and stochastic gradient descent both update parameters using gradients. Describe how their gradients differ and give one practical advantage of each.

Q2. MCQ Pitfall: high LR

Which symptom most suggests the learning rate is too high?

- (A) Training loss decreases smoothly.
- (B) Gradients are never computed.
- (C) The validation set is larger.
- (D) Loss diverges or oscillates strongly.

Q3. True/False

Mini-batch gradient descent often uses noisy gradient estimates compared with full-batch gradient descent.

Q4. Calculation Formula: GD step

One GD step uses $\theta_{t+1} = \theta_t - \eta g_t$. If $\theta_t = 2$, $\eta = 0.1$, $g_t = 3$, compute θ_{t+1} .

Q5. Short Answer

Why can feature scaling speed up local descent methods?

Q6. Short Answer Key: early stopping

What is early stopping and why is it a form of regularisation?

Q7. MCQ

Which optimiser idea uses accumulated past gradients to smooth updates?

- (A) One-hot encoding.
- (B) Momentum.
- (C) Softmax.
- (D) Bootstrap sampling.

Q8. Calculation

A mini-batch size is 64 and there are 6400 training cases. How many mini-batches form one epoch?

Q9. Long Question

A neural network trains slowly and validation accuracy is unstable. Give four training diagnostics or changes you would try.

Q10. Long Question Warning: NaN diagnosis

A neural network's loss becomes NaN after a few mini-batches. Diagnose likely optimisation causes and list four concrete fixes.



W6 Nonlinear Modelling

Q1. MCQ Semester 1 2024 Final (modified)

A spline fit to house prices as a function of floor area follows many tiny oscillations in the training data and performs poorly on cross-validation. Which adjustment is most appropriate?

- (A) Increase the smoothing penalty or reduce effective degrees of freedom.
- (B) Add more knots and force interpolation.
- (C) Replace validation with training error.
- (D) Use a softmax output layer.

Q2. True/False

A GAM with only additive smooth terms automatically captures all interactions among predictors.

Q3. Short Answer Pitfall: additive only

Explain why a high-degree global polynomial can behave poorly near data boundaries.

Q4. Calculation

A cubic spline with more knots is introduced. Predict the qualitative effect on bias and variance.

Q5. MCQ Warning: variance rises

Which model is most interpretable for separate nonlinear effects of age and income?

- (A) Raw-pixel CNN.
- (B) Bidirectional RNN.
- (C) GAM.
- (D) Random labels.

Q6. Short Answer

What does the smoothing penalty in a smoothing spline penalise?

Q7. Calculation Key: smoothness penalty

If validation error increases when degrees of freedom rise from 5 to 20, which df is preferable and why?

Q8. Table/Diagram

Compare basis functions, regression splines and smoothing splines in one compact table.

Q9. Long Question

A pricing team wants nonlinear but explainable effects. Propose a GAM workflow and how you would validate it.

Q10. Long Question Strategy: interpretability trade-off

Compare a GAM and a boosted tree model for nonlinear pricing. When would each be preferable?



W7 Decision Trees and Random Forests

Q1. Short Answer Semester 1 2024 Final (modified)

Explain how bagging with decision trees can reduce prediction variance. Why is the effect usually stronger for the unstable model such as logistic regression?

Q2. MCQ Key: decorrelation

What is the main purpose of random feature selection at RF splits?

- (A) Make every tree identical.
- (B) Remove bootstrapping.
- (C) Force linearity.
- (D) Decorrelate trees.

Q3. True/False

A single fully grown decision tree usually has low variance.

Q4. Calculation Formula: Gini

A node has class proportions 0.7 and 0.3. Compute Gini impurity.

Q5. Short Answer

What is out-of-bag error in bagging?

Q6. Short Answer Pitfall: unstable tree

Why can a decision tree capture interactions without explicit interaction features?

Q7. MCQ

Which hyperparameter directly limits tree complexity?

- (A) Number of pixels.
- (B) Maximum depth.
- (C) Output activation.
- (D) Batch size only.

Q8. Table/Diagram

State two pros and two cons of decision trees for business analytics.

Q9. Long Question

A fraud tree performs perfectly on training but poorly on validation. Diagnose and propose a tree/forest remedy.

Q10. Long Question Strategy: bagging

A single decision tree and a random forest are trained for fraud detection. Compare pruning the tree with using a random forest as overfit controls.



W8 Gradient Boosting

Q1. Short Answer Semester 1 2023 Final (modified)

Which features of XGBoost make it a practical implementation of gradient boosting for tabular business data? Mention three, including one regularisation or efficiency feature.

Q2. MCQ Key: gradient residuals

In gradient boosting, each new tree is usually fitted to what?

- (A) Raw feature names.
- (B) Future test labels.
- (C) Random forest votes only.
- (D) Negative gradients/residual-like errors.

Q3. True/False

A smaller learning rate in boosting often requires more trees.

Q4. Calculation Formula: shrinkage

If $F_0(x) = 10$, tree prediction $h_1(x) = 4$, and shrinkage $\nu = 0.1$, compute $F_1(x)$.

Q5. Short Answer

Why can early stopping help boosted trees?

Q6. Short Answer

Contrast bagging and boosting in one or two sentences.

Q7. MCQ Pitfall: too many trees

Which pair best controls overfitting in gradient boosting?

- (A) Larger learning rate and no validation.
- (B) Shrinkage and validation early stopping.
- (C) Max pooling and softmax.
- (D) Future leakage and deeper trees.

Q8. Calculation

A model with 1000 deep boosting trees outperforms training but not validation. Name two hyperparameters to reduce complexity.

Q9. Long Question

For tabular credit scoring, explain why gradient boosting may outperform logistic regression and what trade-offs it introduces.

Q10. Long Question Strategy: tune stopping

A gradient boosting model keeps improving training loss while validation AUC falls. Explain the diagnosis and a tuning response.



W9 Deep Feedforward Networks

Q1. MCQ Semester 1 2024 Final (modified)

Does a ReLU hidden unit output only values between 0 and 1?

- (A) Yes, it is a probability.
- (B) No, it outputs zero for negative inputs and the input itself for positive inputs.
- (C) Yes, because it is a softmax.
- (D) No, because it always outputs negative values.

Q2. MCQ Pitfall: not softmax

A transaction can have several independent risk flags. Which output design fits?

- (A) One softmax over flags.
- (B) Linear output with no loss.
- (C) Max pooling output.
- (D) One sigmoid per flag with binary cross-entropy.

Q3. True/False

Stacking only linear layers without nonlinear activations is equivalent to one linear transformation.

Q4. Calculation Formula: dense params

A dense layer maps 20 inputs to 8 hidden units with biases. Count parameters.

Q5. Short Answer

Explain the dying ReLU problem and name one mitigation.

Q6. Short Answer Pitfall: dead ReLU

Why does dropout reduce overfitting?

Q7. MCQ

Which activation is standard for mutually exclusive multi-class probabilities?

- (A) ReLU.
- (B) Softmax.
- (C) Max pooling.
- (D) Bootstrap.

Q8. Table/Diagram

Map binary, multi-class, multi-label and regression tasks to output activation and loss.

Q9. Long Question

A deep feedforward net has high training accuracy but poor validation. Propose four changes and explain the trade-off.

Q10. Long Question Key: independent sigmoids

A multi-label risk network is mistakenly built with a softmax output. Explain the error and the correct architecture/loss.



W10 Neural Networks for Images Part 1

Q1. Calculation Semester 1 2024 Final (modified)

A CNN receives a 5×5 single-channel image and applies a 3×3 convolution with padding 1 and stride 1. Compute the spatial size and explain the role of padding.

Q2. MCQ Key: weight sharing

Why do CNNs usually have fewer parameters than fully connected image models?

- (A) They have no weights.
- (B) They ignore channels.
- (C) They use future labels.
- (D) Local connectivity and weight sharing.

Q3. True/False

The number of parameters in a convolutional layer depends on image height and width.

Q4. Calculation Formula: conv size

A 64×64 image uses 5×5 conv, padding 2, stride 2. Compute output size.

Q5. Short Answer

What is the role of padding in early convolutional layers?

Q6. Calculation Pitfall: channel count

Count parameters for 32 filters of size 3×3 on RGB input, with bias.

Q7. MCQ

What does max pooling usually do?

- (A) Learns a weight for every class.
- (B) Down-samples while keeping strong local activations.
- (C) Converts regression to classification.
- (D) Guarantees no information loss.

Q8. Table/Diagram

Sketch the tensor shapes for Input $128 \times 128 \times 3 \rightarrow \text{Conv}32 \text{ k5 p2 s1} \rightarrow \text{Pool}2 \text{ s2}$.

Q9. Long Question

Explain why stride 2 can hurt small-object recognition and how padding or architecture choices can mitigate this.

Q10. Long Question Strategy: track HWC

For Input $128 \times 128 \times 3 \rightarrow \text{Conv}(32, 5 \times 5, \text{pad } 2, \text{stride } 1) \rightarrow \text{MaxPool}(2 \times 2, \text{stride } 2)$, compute the tensor shapes and first-conv parameter count.



W11 Neural Networks for Images Part 2

Q1. Long Question Semester 1 2024 Final (modified)

A CNN for small product-defect images loses accuracy after aggressive pooling and stride. Explain the trade-off of pooling/stride and propose two architecture or training changes that could improve generalisation without discarding detail.

Q2. MCQ Key: 1x1 channel mix

What is a common reason for a 1×1 convolution?

- (A) Increase spatial kernel to 7×7 .
- (B) Perform recurrent memory.
- (C) Remove all nonlinearities.
- (D) Mix channels and change channel dimension.

Q3. True/False

Batch normalisation replaces the need for a correct output activation and loss.

Q4. Short Answer Strategy: residual path

How do residual connections help train very deep CNNs?

Q5. Calculation

A conv block outputs $16 \times 16 \times 64$. Global average pooling produces what vector length?

Q6. Short Answer

Compare label smoothing and mixup.

Q7. MCQ Pitfall: mixup labels

Which technique creates synthetic examples by convexly combining examples and labels?

- (A) OOB error.
- (B) Mixup.
- (C) Cost-complexity pruning.
- (D) Softmax only.

Q8. Table/Diagram

List two benefits and two risks of data augmentation for image tasks.

Q9. Long Question

Design two improvements to a simple Fashion-MNIST CNN baseline and justify why each may improve test accuracy.

Q10. Long Question Warning: image validation

A shelf-image CNN overfits a small labelled dataset. Propose two architecture/training improvements and one validation check.



W12 Neural Networks for Sequences Part 1

Q1. Short Answer Semester 1 2024 Final (modified)

In an RNN, explain the role of the hidden state and why learning long-term dependencies can be difficult for a vanilla network.

Q2. MCQ

Sentiment classification of a complete review is which sequence formulation?

- (A) One-to-many.
- (B) Image segmentation.
- (C) Unsupervised clustering only.
- (D) Many-to-one.

Q3. True/False Pitfall: future leak

A bidirectional RNN is always safe for forecasting deployment.

Q4. Short Answer

A model uses 10 time steps with 6 features each and returns one churn probability. Name the input and output structure.

Q5. Short Answer Key: embeddings

What is an embedding layer used for in text models?

Q6. Short Answer

Why can vanilla RNN gradients vanish over long sequences?

Q7. MCQ Strategy: time split

Which validation split is safest for next-month sales forecasting?

- (A) Randomly mix future months into training.
- (B) Train on earlier months, validate on later months.
- (C) Tune on the final test month repeatedly.
- (D) Use target values as inputs.

Q8. Table/Diagram

Match one-to-many, many-to-one and many-to-many to an example.

Q9. Long Question

A support centre wants transcript-level sentiment. Describe preprocessing, model output and one leakage risk.

Q10. Long Question Warning: causal inputs

A next-week sales model uses a bidirectional RNN over a window that includes future weeks during validation. Explain the problem and redesign the split/input.



W13 Neural Networks for Sequences Part 2

Q1. Short Answer Semester 1 2024 Final (modified)

Explain how the update and reset gates in a GRU affect information flow through time. Relate your answer to a churn sequence where old behaviour may or may not remain relevant.

Q2. MCQ Key: reset gate

In a GRU, what does the reset gate mainly control?

- (A) The final class count.
- (B) Spatial pooling.
- (C) The number of convolution filters.
- (D) How much previous hidden state enters the candidate state.

Q3. True/False

LSTM and GRU gates are designed partly to improve information flow over longer sequences.

Q4. Short Answer Formula: update retention

If a GRU update gate strongly retains the old state, what qualitative effect does this have on h_t ?

Q5. Short Answer

Compare GRU and LSTM in terms of gates and parameters.

Q6. Short Answer Pitfall: rare churn

Why might class weights or threshold tuning be important in churn prediction?

Q7. MCQ

Which design avoids future leakage in rolling churn prediction?

- (A) Use next month's cancellation flag as a feature.
- (B) Use only observations up to prediction date.
- (C) Use bidirectional context beyond the prediction date.
- (D) Randomly copy labels into inputs.

Q8. Table/Diagram

Summarise reset gate, update gate and candidate state in one compact table.

Q9. Long Question

A sequence model performs well offline but poorly after launch. Give three possible causes and checks.

Q10. Long Question Strategy: GRU vs LSTM

For churn sequences, compare GRU and LSTM and explain how you would handle class imbalance in evaluation.



Answer Key

W1 Introduction to Machine Learning

- Q1.** Two valid safeguards include deployment-like validation/stress testing and risk-controlled decision rules. tail-error analysis, uncertainty intervals, human review, exposure caps, drift monitoring, and explaining why average prediction accuracy alone is insufficient for high-value automated purchases.
- Q2.** D. Supervised learning uses labelled examples.
- Q3.** False. It also depends on representative validation, no leakage, and stable deployment distribution.
- Q4.** Structured: customer table with tree/GLM/boosting. Unstructured: images/text with CNN/RNN/embedding model.
- Q5.** Extra FP cost 600; FN savings 1000; net cost decreases by 400.
- Q6.** It sets a sanity benchmark, exposes leakage/data issues, and clarifies whether complexity is worth the cost.
- Q7.** B. It specifies unit, label, and horizon.
- Q8.** Regression: continuous value; binary: probability/class for two classes; multi-class: one class among K.
- Q9.** Data quality/bias, validation/leakage, calibration/uncertainty, explainability/regulation, monitoring and human review.
- Q10.** Target should be observable and time-bound; validation should mimic deployment and include tail/cost analysis; uncertainty can trigger human review or conservative thresholds; monitoring should track drift, calibration and costly errors.

W2 Machine Learning Fundamentals

- Q1.** A. The pattern is overfitting: training error is much lower than validation error, so the model has likely learned sample-specific noise.
- Q2.** Second model likely generalises better; first overfits.
- Q3.** A. Accuracy can be high by predicting the majority class.
- Q4.** False. It penalises large coefficients and usually reduces variance.
- Q5.** Parameters learned from data e.g. coefficients/weights; hyperparameters set/tuned e.g. lambda, depth, learning rate.
- Q6.** Accuracy $960/1000=96\%$
- Q7.** When probability quality/calibration matters or threshold may change.
- Q8.** C.
- Q9.** Use validation probabilities, contact cost and expected benefit, choose threshold maximising expected profit or satisfying precision/recall constraints.
- Q10.** Class imbalance allows high accuracy from majority predictions. Use recall/precision/F1/PR-AUC, choose threshold using fraud loss and investigation capacity, and report confusion matrix at that threshold.

W3 Machine Learning in Practice

- Q1.** Accuracy can hide failure on the minority class. Strong answers mention precision/recall/F1, ROC-AUC or PR-AUC, balanced accuracy, log loss/calibration, stratified or repeated CV, and choosing thresholds by business cost.
- Q2.** D. Otherwise validation/test information leaks.
- Q3.** True.
- Q4.** Generated after prediction time; directly encodes target/outcome; available only after decision; duplicates label.
- Q5.** Mean = 0.76.
- Q6.** Each fold must estimate preprocessing using training folds only to avoid leakage.
- Q7.** B.
- Q8.** High bias: high train/val error; high variance: low train high val error; shift: validation good but deployment poor.
- Q9.** Use time-based or customer-separated split matching deployment; avoid future data and same-customer leakage.
- Q10.** Use time-aware and/or customer-grouped splits, fit preprocessing inside folds, tune on validation only, reserve a final test period, and compare models under the same metric and horizon.

W4 Training Machine Learning Models Part 1

- Q1.** Regularisation penalises overly large coefficients to control variance and improve generalisation. Larger strength usually increases bias, lowers variance, stabilises estimates, and may shrink or remove weak predictors depending on penalty.
- Q2.** D.
- Q3.** True.



Q4. $\sigma(0) = 0.5$.

Q5. It improves conditioning and makes penalties comparable across coefficients.

Q6. L1 can set coefficients exactly to zero; L2 shrinks but usually keeps all.

Q7. B.

Q8. $-\log(0.8)$.

Q9. Interpretable coefficients/probabilities, regularisation controls overfit; limitation: misses complex nonlinear interactions.

Q10. Prefer the model with better validation log loss/calibration and acceptable interpretability. Discuss penalty strength, coefficient stability, bias-variance trade-off, and threshold choice after probability estimation.

W5 Training Machine Learning Models Part 2

Q1. Batch gradient descent uses the full training set for each gradient, giving stable but expensive updates. SGD uses one observation or a small random batch, giving cheaper noisy updates that can scale to large data and sometimes escape shallow poor regions.

Q2. D.

Q3. True.

Q4. 1.7.

Q5. It reduces elongated contours and makes a single learning rate work across dimensions.

Q6. Stop when validation stops improving; prevents fitting later noise.

Q7. B.

Q8. 100.

Q9. Check scaling, LR schedule, batch size, optimiser, activation, regularisation, train/val curves, class balance.

Q10. Likely high learning rate/exploding gradients/bad scaling/numerical instability. Fix by lowering LR, scaling inputs, gradient clipping, stable loss, better initialisation, smaller batch or optimiser/schedule changes.

W6 Nonlinear Modelling

Q1. A. Increase smoothness or reduce effective degrees of freedom. The curve is fitting noise rather than signal, so validation should favour a less wiggly spline.

Q2. False.

Q3. It can oscillate and is influenced globally by local data changes.

Q4. Bias tends to fall; variance tends to rise.

Q5. C.

Q6. Curvature/wigginess, often integrated squared second derivative.

Q7. Prefer 5 unless other evidence; 20 likely overfits.

Q8. Basis expansion creates transformed predictors; regression spline uses knots; smoothing spline chooses smoothness via penalty.

Q9. Fit train-only preprocessing, smooth terms, tune smoothness by CV, inspect partial effects/residuals, compare baseline and holdout.

Q10. GAM is preferable for interpretable smooth marginal effects and simpler reporting; boosted trees are preferable for stronger prediction with interactions. Validate both and weigh accuracy against explanation needs.

W7 Decision Trees and Random Forests

Q1. Bagging trains trees on bootstrap samples and averages/votes their predictions. Deep trees are unstable, so averaging many decorrelated trees reduces variance; stable learners such as logistic regression change less across bootstrap samples, so bagging helps less.

Q2. D.

Q3. False.

Q4. $1 - 0.49 - 0.09 = 0.42$.

Q5. Prediction for each observation using trees whose bootstrap sample did not include that observation.

Q6. Splits are conditional on previous splits, creating rule interactions.

Q7. B.

Q8. Pros: interpretable rules, nonlinear/interactions. Cons: unstable, overfit, greedy.



Q9. Overfitting; prune/depth/min leaf, bagging/RF, class metrics, validation tuning.

Q10. Pruning simplifies one tree and improves interpretability while raising bias. Random forest averages decorrelate, reduce variance and often improves accuracy but loses simple rule-level transparency.

W8 Gradient Boosting

Q1. Strong answers mention scalable tree construction, shrinkage, depth/leaf regularisation, column/row subsampling, missing-value handling, custom objectives/metrics, parallelism, and strong performance on nonlinear tabular interactions.

Q2. D.

Q3. True.

Q4. 10.4.

Q5. It stops before later trees fit validation noise.

Q6. Bagging averages parallel bootstrap models to reduce variance; boosting sequentially corrects errors to reduce bias and improve fit.

Q7. B.

Q8. Lower depth/leaves; stronger regularisation; subsampling; lower LR plus early stopping.

Q9. Captures nonlinearities/interactions, strong predictive power; less interpretable, tuning/runtime, overfit risk.

Q10. It is overfitting later trees. Use early stopping, smaller learning rate with validation-selected iterations, shallower trees, subsampling, stronger regularisation, and metric aligned with the business problem.

W9 Deep Feedforward Networks

Q1. B. ReLU is zero for negative inputs and equals the input for positive inputs, so it is unbounded above and is not a probability output.

Q2. D.

Q3. True.

Q4. $20 \cdot 8 + 8 = 168$.

Q5. Unit stuck with negative preactivation gives zero gradient; use lower LR, leaky ReLU/ELU, good init, BN.

Q6. It injects noise by dropping units, discouraging co-adaptation.

Q7. B.

Q8. Binary sigmoid/BCE; multi-class softmax/CE; multi-label sigmoids/BCE; regression linear/MSE or MAE.

Q9. Dropout, weight decay, smaller model, early stopping, more data/augmentation, BN; trade-off bias/runtime.

Q10. Softmax forces labels to compete and sum to one, which is wrong for non-exclusive flags. Use one sigmoid per label with binary cross-entropy summed/averaged across labels.

W10 Neural Networks for Images Part 1

Q1. With padding 1 and stride 1, output size is 5 by 5. Padding lets the filter be centred near boundary pixels and preserves spatial size in this case.

Q2. D.

Q3. False. It depends on kernel size, input channels, output channels and biases.

Q4. $\lfloor (64 + 4 - 5)/2 \rfloor + 1 = 32$, so 32 by 32.

Q5. Preserve spatial size and allow filters to use boundary pixels.

Q6. $32 \cdot (3 \cdot 3 \cdot 3 + 1) = 896$.

Q7. B.

Q8. $128 \times 128 \times 3 \rightarrow 128 \times 128 \times 32 \rightarrow 64 \times 64 \times 32$.

Q9. Stride samples fewer locations; preserve resolution early, use padding, smaller stride, multi-scale features, later pooling.

Q10. Conv output is $128 \times 128 \times 32$; pooling gives $64 \times 64 \times 32$; first conv parameters are $32(5 \cdot 5 \cdot 3 + 1) = 2432$.

W11 Neural Networks for Images Part 2

Q1. Pooling/stride reduce computation and add some translation robustness but may discard small local evidence. Improvements include reducing early stride, delaying pooling, using padding, adding convolutional capacity, batch normalisation, augmentation, or skip/multi-scale features.



- Q2. D.
- Q3. False.
- Q4. They provide shortcut paths for gradients and allow layers to learn refinements.
- Q5. 64.
- Q6. Label smoothing softens labels; mixup forms convex input and label combinations.
- Q7. B.
- Q8. Benefits: invariance, larger effective data. Risks: unrealistic images, label distortion.
- Q9. Add conv capacity/BN/dropout/augmentation/LR schedule; explain representation, stability, regularisation.
- Q10. Valid choices include transfer learning, augmentation, dropout, weight decay, batch norm, smaller classifier or early stopping. Validate on held-out stores/time periods and inspect confusion/error cases.

W12 Neural Networks for Sequences Part 1

- Q1. The hidden state summarises information from previous time steps. Vanilla RNNs struggle with long dependencies because gradients are repeatedly multiplied through time, causing vanishing/exploding gradients.
- Q2. D.
- Q3. False. It may use future information unavailable at prediction time.
- Q4. Sequence input length 10 with 6 features; many-to-one binary output.
- Q5. It maps discrete tokens to dense learned vectors.
- Q6. Backprop multiplies many recurrent Jacobians, shrinking gradients.
- Q7. B.
- Q8. Caption generation; sentiment/churn; sequence tagging/translation.
- Q9. Tokenise/pad/embed, RNN final/pooled state to softmax; leakage from post-outcome notes or future messages.
- Q10. Future weeks create information leakage and optimistic performance. Use only observations available up to prediction date, causal windows, and train-on-past validate-on-later time splits.

W13 Neural Networks for Sequences Part 2

- Q1. The reset gate controls how much previous hidden state contributes to the candidate state; the update gate controls how much old state is retained versus replaced. For churn, gates help keep persistent behaviour while resetting stale history after regime changes.
- Q2. D.
- Q3. True.
- Q4. The new hidden state stays close to the previous hidden state.
- Q5. GRU has reset/update and fewer parameters; LSTM has input/forget/output gates and cell state.
- Q6. Churn is often rare; accuracy may hide missed churners.
- Q7. B.
- Q8. Reset filters past for candidate; update controls retention; candidate proposes new content.
- Q9. Temporal shift, leakage, entity split issue, class imbalance, calibration drift; check time-based validation and monitoring.
- Q10. Both use gates for long dependencies; GRU is simpler with reset/update gates, LSTM has cell state plus input/forget/output gates. Use PR-AUC, recall/precision, class weights or threshold tuning, not accuracy alone.